Automotive Technical Service Diagnostic Bulletin

Document ID: DIAG-C-CLASS-P0301
Classification: OEM Level 3 Diagnostic & Repair Procedure
Target Vehicle: Mercedes-Benz C-Class (W206 / C 300)
Powertrain: 2.0L Turbocharged Inline-4 (M254) with 2nd Gen ISG (Beltless 48V Mild-Hybrid)
Engine Control System: Bosch MD1CS006 (Engine Control Module N3/10) - High-Speed CAN-E2 Architecture
Electrical Architecture: Integrated 12V / 48V Architecture with 48V DC/DC Converter (N83/1)
Fuel Injection Architecture: Single Cam-Driven High Pressure Pump (HPP) with Solenoid Direct Injectors (200-250 bar)
Firing Order: 1-3-4-2

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1. DIAGNOSTIC TROUBLE CODE (DTC) SPECIFICATION

- **DTC Code:** `P0301`
- **Standard Fault Description:** Cylinder 1 Misfire Detected
- **System Definition:** Combustion failure or poor flame propagation specifically localized to Cylinder 1 based on CKP sector timing delta.
- **MIL (Malfunction Indicator Lamp) Status:** Illuminated (Continuous or Flashing depending on catalyst severity)
- **Monitoring Condition:** Continuous monitoring during engine running cycle with coolant temp > 60°C and vehicle speed > 0 km/h.

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2. PLATFORM-SPECIFIC ROOT CAUSE ANALYSIS (RCA)

Root cause analysis customized specifically for the **Mercedes-Benz C-Class (W206 / C 300)**:

1. **Solenoid Fuel Injector Y76/1 Nozzle Tip Coking:** Lacquer formation on 5-hole multi-hole tip due to heat soak-back on compact M254 cylinder head.
2. **Compact Pencil Coil T1/1 Internal Short:** High-voltage breakdown in secondary winding causing intermittent spark extinction under boost.
3. **Exhaust Valve Carbon Accumulation Cylinder 1:** Variable Camtronic exhaust lift actuator sticking in partial lift profile.

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3. DEDICATED WIRING DIAGRAM & CIRCUIT PINOUT

Schematic configuration and electrical pin connections specifically for the C-Class control harness:

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[N3/10 ECM Connector 'K']                  [Solenoid Injector Y76/1]
Pin 72 (Cyl 1 Injector Control +) ----> Pin 1 (Peak & Hold +12V/65V)
Pin 73 (Cyl 1 Injector Ground Drive) -> Pin 2 (ECM Low-Side Switched)

[N3/10 ECM Connector 'K']                  [Ignition Coil T1/1]
Pin 21 (5V Trigger) ---------> Pin 3 (Ignition Command)
Circuit 15 (Switched Ignition 12V) -------> Pin 1 (Power Supply)
Ground W10 (Engine Block Ground) ---------> Pin 2 (Power Return)
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- Injector Coil Resistance: 1.4Ω - 2.1Ω at 20°C.

Electrical Reference Points

- **Primary Ground Reference:** `W10 (Engine Block Ground Point, Right) & W9 (Front Wheel Arch Ground)`
- **Power Distribution Node:** `Engine Bay Prefuse Box F32/1 & Front SAM Module (N10/1)`

- **Signal Logic Voltage Range:** 0.0V to 5.0V regulated DC (unless high-voltage direct injection rail)

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4. STEP-BY-STEP DIAGNOSTIC & VERIFICATION PROCEDURE

Follow this OEM workshop sequence to isolate the failure on this platform:

1. Swap Pencil Ignition Coil T1/1 to Cylinder 2; road test with XENTRY live counter.
2. Measure resistance of Solenoid Injector Y76/1 across Pin 1 and Pin 2 (Spec: 1.8Ω ± 0.2Ω).
3. Perform Camtronic actuator cylinder balance test to ensure exhaust rocker arms switch properly between low and high lift.
4. Remove spark plug 1 and check for fuel fouling or porcelain hairline cracks.

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5. PRECISE RESOLUTION & REPAIR PROCEDURE

- Replace Solenoid High-Pressure Fuel Injector Y76/1 (Part # A254070xxxx) with new copper combustion seal washer.
- Calibrate cylinder 1 injector flow quantity offset in ECM adaptation menu.
- Replace set of 4 ignition coils if secondary winding breakdown is evident (prevent cascading failure on M254).

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6. POST-REPAIR QUALITY VALIDATION & RE-TESTING

1. Clear all DTCs from Engine Control Module (`Bosch`) and secondary gateway modules.
2. Perform comprehensive drive cycle:
3. Cold start idle for 3 minutes (monitors secondary air / catalyst preheat and cold misfire).
4. Accelerate smoothly to 50 km/h and hold for 2 minutes.
5. Cruise at 80–100 km/h on highway for 10 minutes (validates downstream O2 sensor and fuel trim closed-loop balance).
6. Coasting deceleration (fuel cut-off check) for 5 seconds without braking.
7. Re-scan control units with diagnostic tool; confirm all OBD-II readiness monitors show **READY** / **COMPLETE** and no pending DTCs are stored.